



Original research article

Developing carbon dioxide removal policy and anticipatory perspectives in the United Kingdom and United States

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ABSTRACT

The deployment of Carbon Dioxide Removal (CDR) to GtCO₂ scale by the mid-century is integral to realising net zero. Some CDR methods are technically maturing whilst some are in the early stages of innovation. At the same time, first mover nations are implementing policy and regulatory frameworks to establish this sector. Policy analysis around CDR has therefore become an area of focus as a function of its importance in establishing, developing and scaling a just, equitable and sustainable net zero transition. Policy development for sustainable transitions tends to use consolidative modelling approaches. These, often protracted, analyses can become quickly outdated in dynamic fast-moving sectors, and be unable to manage the multiple objectives that actors seek when innovating. The innovation rate required to scale the CDR sector and the extent of uncertainty to realise net zero, questions the applicability of these approaches.

This research proposes a novel rapid policy analysis process, based on the decision theatre format, that uses strategic foresight to develop anticipatory, values-based CDR policy insights. This contribution prototypes this rapid approach on two pioneer national CDR developers - the United Kingdom and United States. Through its application, the method identifies opportunities for globally collaborative CDR policy intervention - a roadmap, a coordination hub, and agile policy processes - supported by regional policy specialisations - coordinating offices, measurement standards, and public capacity building - to achieve GtCO₂ scale CDR deployment. Most importantly, the proposed approach provides a rapid method for sustainable transitions policy analysis in contexts of deep uncertainty.

1. Introduction

The Intergovernmental Panel on Climate Change (IPCC) recognises Carbon Dioxide Removal (CDR) as unavoidable to realise net zero [1]. Mitigation through emissions reductions alone will fail to limit global temperature rise in line with climate targets [2]. CDR is necessary to address hard-to-abate residual emissions. This will require a scale-up of CDR from its current nascent state, whereby technological deployment is immature, catalytic policies do not yet exist and markets are only slowly emerging, in order to realise a global GtCO₂ scale sector in less than 30 years [3]. The extent of uncertainty within which a CDR sector is being developed can therefore be considered deep [4].

The toolbox of potential CDR methods and technologies has expanded in recent years. Focus is no longer solely directed towards nature-based removal, such as Afforestation, and most new methods

include a form of engineered intervention. Scientific assessments [5,6,7,8] increasingly indicate these methods are technologically maturing. Simultaneously developers are beginning to emerge as they seek to generate a potentially climate relevant CDR pipeline.

Despite technological progress, deployment of engineered CDR remains limited. For Direct Air Capture (DAC), for example, (at the time of writing) the only operational plant has a removal capacity of 4 ktCO₂ per annum (pa) [9]. Similarly, Bio-oil and Biochar projects are deploying, but again at smaller scale, cumulatively removing in the order of ktCO₂ pa. Projections suggest a scale-up of engineered methods by a factor of 1,300 is required by the mid-century to achieve GtCO₂ scale and meet climate targets [10]. The gap between technical maturity and present deployment demonstrates that other dimensions of CDR scaling, beyond technological maturity, are required to support CDR emergence – namely socio-political aspects [11,12].

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The generation of anticipatory, macroeconomic climate policies is essential to overcome the present innovation gap. However, due to the nascent nature of this sector, CDR policy is currently limited [13]. Effective policy insight generation and implementation will be of importance for successful sector scaling.

Traditional insight generation tools, that are adopted by policymakers in sustainable transitions, often use consolidative modelling methods. These methods, such as Integrated Assessment Models (IAMs), gather all relevant knowledge into a single package which, once validated, are used as a surrogate for the real world [14]. These models are based on an economic rational optimisation approach where it is assumed users have access to perfect information. They are designed to meet a future climate target by outputting insights based on existing and past information, which makes them backwards looking. For the CDR sector, which is nascent and subject to deep uncertainty, parameterising these models is challenging and the analysis timeframe can be protracted.

Decisions for scaling this sector that are informed by these methods will likely be limited. Investment decisions are not made based on economically rational optimised information. Rather they are made by individuals in firms and are reflexive to other actors operating within the CDR sector and across the broader economy [15]. Thus, insights generated by traditional methods are unlikely to reflect business decisions and will become quickly outdated in a fast-moving sector. CDR calls for a more strategic, values-based and exploratory approach to insight generation.

To address the rapid rate at which the CDR sector needs to evolve, this research prototypes a novel rapid policy analysis process. This process, based on a decision theatre format [16] within a strategic foresight approach [17], seeks to provide tenets of anticipatory regulation [18] to the CDR policy landscape. The ability to accommodate forward looking insight will be fundamental to achieve critical mass in what will need to be a global CDR sector.

The literature on CDR policymaking and CDR policy design is growing [6,7,12,13,19,20,21]. Notably, Honegger et al. [20] outlined key governing principles required for CDR policy generation and Schenit et al. [13] mapped the fragmented regional CDR policy landscape that is emerging. By building on these works, this research presents a first of its kind exploration of the role of two first mover nations – the United Kingdom (UK) and United States (US) – in deploying policies and initiatives to stimulate the sector as part of their transition to net zero.

With momentum for a global CDR sector accelerating, some nations are developing preliminary strategies for establishing this sector, including Canada, China, the European Union, Norway, the UK and US. In the last decade, the UK and US have been particularly active in this landscape, establishing supporting initiatives which combined are valued at over US\$10 billion, to develop CDR. In this research, the extent and scale of these initiatives are mapped and analysed with insights unpacked using this novel, exploratory approach.

The aim is to establish a replicable process for undertaking policy analysis in net zero transitions. The rapid policy analysis process examines early developments in the UK and US to identify opportunities for policy collaboration internationally and opportunities for specialisation nationally. This in turn is used to align current fragmented policy initiatives and reconcile technical development and deployment, which is commensurate with realising net zero.

In Section 2, the current gap in methodological approaches for policy insight generation is explained. In Section 3, how the rapid policy analysis process is applied to guide this research on the establishment of a CDR sector in the UK and US is reported. Section 4 presents the policy insights identified by applying this novel process. Section 5 reflects on the possible applications of these insights as the CDR sector develops into a global sector, identifies further research opportunities, and invites others to use this process, refine it, critique it and apply it to other sectors within the net zero transition.

2. The need for a rapid policy analysis process

2.1. Consolidative modelling methods

Traditional decision support tools adopted by policymakers to generate policy insights tend to be based on consolidative modelling approaches [22]. Consolidative models gather all relevant knowledge into a single package which project support needs based on pre-defined outcomes. These models, appropriate for describing complicated systems, are frequently misapplied to design climate policy around net zero [14]. For example, the consolidative IAMs used by the IPCC at the international scale, rather than being used as a decision support tool [23], have effectively become the prescriptive organising framework [13] for international climate policy design.

Consolidative models are also used at the regional scale, for example in modelling for regional energy system transitions [24]. Different countries have customised energy models, such as the National Grid's Future Energy Scenarios [25] in the UK or the Department of Energy's (DOE) National Energy Modeling System (NEMS) [26] in the US, the outcomes of which policymakers utilise to generate policy. The shortcomings of the consolidative modelling methods for the UK energy system transition regionally have been examined in Hall et al. [17] and presented in Table 1. Limitations identified for the UK model have key parallels to energy systems models that are used in other regions [27].

The reliance on consolidative models, whether international IAMs or regional energy systems models, has a number of limitations. In using an economic optimisation framework with the assumption of perfect information for pre-defined goals using pre-defined parameters,

Table 1
UK Energy System Modelling policy analysis tools - features and limitations.

Model name and analytical approach	Features	Limitations
MARKAL Bottom-up: Econometric - Cost Optimisation [28]	• High level of technological detail and sectoral coverage • Perfect foresight of competitive markets	• Lack macroeconomic completeness and fails to capture realistic micro-economic interactions
MARKAL – MACRO Hybrid: Econometric - Cost optimisation and partial equilibrium [28]	• Combined the detail of MARKAL with macroeconomic trends • Uses non-linear programming	• Decisions are based on solely economic benefit • Does not take into account the behaviour of individual actors and customers • Assumes that decisions are based solely on economic benefit • Does not take into consideration other socio-technical interactions i.e. customers • Lack of heterogeneity in demand • Oversimplification of technologies • Coarse representation of time steps • Endogenous or exogenous factors which affect prices of technology are absent • Assumes decisions are based on economic benefit
TIMES Bottom-up: Econometric - Cost Optimisation [29]	• Four levels of time granularity • Decides which energy carriers and technologies can be used to meet the demand • Data gaps are filled by interpolating or extrapolating data values linearly	• Trade-off between depth and breadth in the 3 domains • High level of complexity in modelling • No insights on business models or consumers
ESME Bottom-up: Econometric - Cost Optimisation [30]	• Technology-rich • Limited heterogeneity on demand-side	
STETs Hybrid: Techno-economic modelling with socio-technical framework [31]	• Covers three domains: • Techno-economic detail • Actor heterogeneity • Transition pathway dynamics	

From Hall et al. [17].

consolidative models are sensitive to economic assumptions, which are often highly uncertain. In their design, they obscure the value judgements implicit in their goals [32] and can blind decision-makers to the robustness of achieving pathways [33]. They can also be time consuming to compile, as a function of the need to input new policy parameters and accommodate constraints to generate the requisite policy insight.

For fast-evolving, nascent sectors - given these shortcomings - such models will likely struggle to generate anticipatory perspectives. The CDR sector, relative to the energy sector, has less available past data and that which is available is subject to greater uncertainty and complexity when parameterising for possible futures [4,34]. In such deeply uncertain circumstances exploratory approaches are more appropriate.

2.2. The rapid policy analysis process

The rapid policy analysis process is a prototype exploratory approach, combining strategic foresight, human centred design and decision theatre methods to not only address the extent of uncertainty in the CDR sector, but also facilitate quicker generation of anticipatory insights. With the speed with which CDR policy development needs to take place, a more rapid approach is needed to ensure policy generation that is responsive to the needs of actors in the sector.

Exploratory approaches map assumptions onto consequences, without privileging any one set of assumptions [35]. The construct embraces processes which engage actors who bring a diversity of priorities, goals and values to the decision, and embraces deep uncertainty to explore decision-making objectives around possible policy design. Exploratory approaches accommodate that some uncertainty and actor behaviours need be addressed within the broader elements involved in the decision-making rather than based on market-based, economically rational allocations which are often opaquely endogenised within a model.

This novel process augments the strategic foresight and human centred design exploratory methods. Strategic foresight is a process in 'futures' work [36] where different future outcomes are used to develop different scenarios. Human centred design is an organising construct for business model innovation, increasingly applied in sustainable transitions [17], that by prioritising human inputs identifies solutions that are: desirable; viable; and feasible [37]. Adapting these socio-technical transitions-based approaches allows for the co-evolutionary development of non-parametric values-based judgements between different actors within the emergent sector, including policymakers. This broadens many of the cultural and differing values-based tensions that take place when innovating in deep uncertainty, allowing for informed policy design rather than masking them artificially within a consolidative model [38]. The rapid policy analysis process sits within the strategic foresight and human centred design approach and augments the feasibility dimension as a means of further unpacking policy insights.

Generating values-based insights within this exploratory approach requires a workshop method for iteration. In this prototype of the rapid policy analysis process the decision theatre is used. Decision theatres were pioneered by Arizona State University to explore areas of climate uncertainty and reveal intricacies of complex systems from engaged and informed individuals [16,39]. The method develops upon the concerns of actors closely aligned to a topic, by creating a discursive narrative defined by informed perspectives. It is assumed that understanding these perspectives within the CDR sector will reveal more targeted priorities and insights at a faster rate than conventional approaches.

Traditional governance mechanisms tend to be prescriptive in nature which can struggle to address deep uncertainty and fast-moving technology development [40]. However, 'anticipatory governance' [18] frameworks are being developed that have a future-facing, inclusive, iterative and experimental dimension to innovation. Governments globally are gradually subscribing to elements of anticipatory governance [41] as a means to better shape support functions for adaptive and

reflexive policy design. The bringing together of strategic foresight, human centred design and decision theatres for anticipatory insights within the rapid policy analysis process is schematised in Fig. 1.

The top half of Fig. 1 projects elements of human centred design as a method for success in business strategies in the co-evolutionary framework [42], designed by Hall et al. [17]. In this framework human values are prioritised during business model innovation. Human centred design is comprised of:

- **Desirability:** the attractiveness of the services and products that are generated by the business model and/or proposition. This is the business-to-business relationships and the willingness for consumers to adopt the services that are being provided.
- **Viability:** the quantification of the revenue streams that are available to business models and actors as a function of macroeconomic policies, grants and subsidies. This assesses the ability for a business model to access revenue streams, capture value and make money which is integral for success. The robustness of those revenue streams will need to be assessed in multiple futures to accommodate uncertainty.

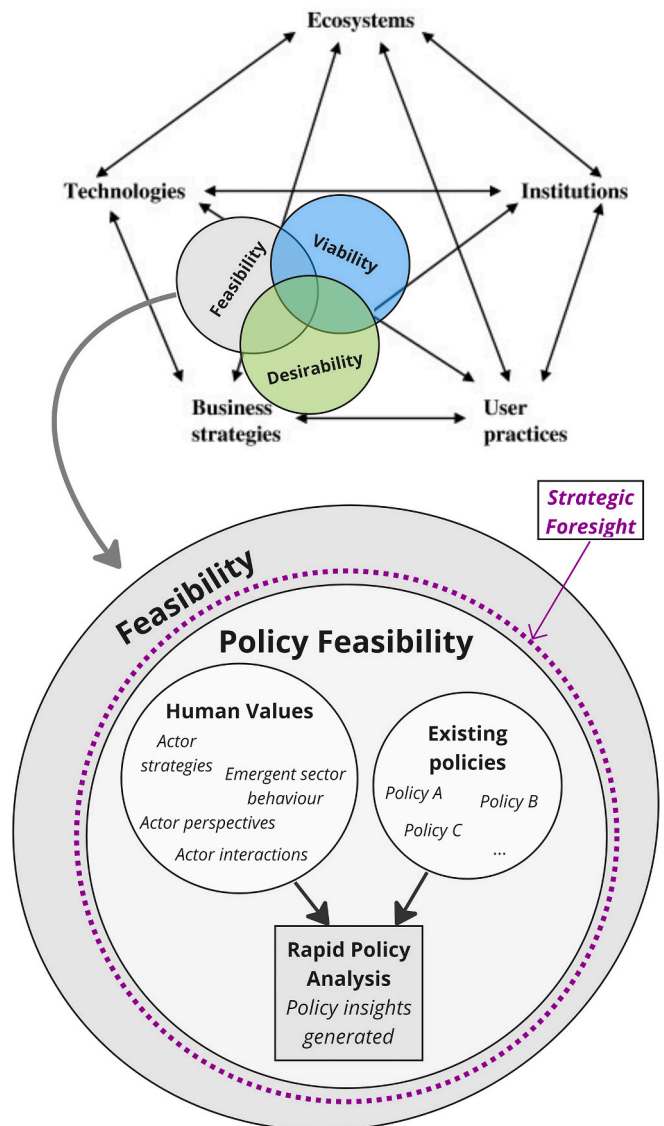


Fig. 1. The rapid policy analysis process as an augmentation to the strategic foresight and human centred design approach developed by Hall et al. [17].

- **Feasibility:** the stress testing of technologies, policies, regulations and standards required to develop successful business models and/or value propositions. This assesses the needs for the business model to be realised not only to ensure both financial and non-financial returns. This includes technology innovation, required regulation, standards and the policy landscape that, when enabled, allow the business model to be realised.

The exploratory, rapid policy analysis process, presented in the bottom half of Fig. 1, builds upon this framework by prioritising values-based insights and the existing policy landscape within a strategic foresight approach, to generate anticipatory insights as a means for policy feasibility. This ethnographic approach moves away from optimised market-based allocations of traditional consolidative modelling methods and allows decisions to be made based on the actor's response to policies.

Section 3 reports how this rapid policy analysis process for policy feasibility has been practically applied to seek anticipatory insights for the emergent UK and US CDR sectors.

3. Method: applying the rapid policy analysis process to the UK and US CDR sectors

3.1. The rapid policy analysis process in practice

Outlined in Fig. 1, this research develops an ethnographic, qualitative method for CDR policy design, that is informed by engaged actors as a means to generate policy feasibility. In this contribution, it does this by examining two regions – the UK and US – to highlight key priorities in CDR establishment. Collecting preliminary perspectives from actors who are regional CDR experts allows for more focused identification of the socio-technical priorities for scale-up, and ultimately reveals the likely direction policy support will act, or needs to act, in the near term. The exploration of the barriers and challenges to CDR deployment also allows the identification of opportunities to overcome these obstacles. Prioritising these opportunities into themes then allows current CDR sector policy design to be built upon.

The rapid policy analysis process is summarised in Fig. 2. In this study the method was conducted in four parts:

1. **Preliminary insights identification:** UK and US CDR insights were identified in two processes.
 - a. **Review of regional CDR policy landscape:** The UK and US were used as case studies for this research as they are first movers in CDR policy development. The existing policy and regulatory mechanisms in both regions were mapped and analysed.
 - b. **Semi-structured interviews:** Interviews were held with sector experts. The purpose of the semi-structured [43] interview was two-fold: (1) to understand individual participant values and the foundations of their perspective; and (2) to establish the national concerns that need further exploration. They act as a preliminary evidence base for the blueprint of the decision theatres. The interviewed CDR experts included 8 policy advisors, 9 project developers, 3 representatives from environmental NGOs, 4 representatives from heavy emitting industries (e.g. oil and gas) and 3 academics, split between the UK and US.
2. **Decision theatres:** Selected participants from the interview cohort were placed in two-hour, regional decision theatre workshops to facilitate and catalyse discursive dialogue between stakeholders of the same region, to further reveal priorities and facilitate the human centred approach. These workshops assigned participants specific personas - caricatures of actors in the CDR ecosystem - to discuss responses to hypothetical CDR developments over multiple timeframes. This step is expanded upon in Section 3.2.
3. **Thematic analysis:** Interviews and workshops revealed a dataset of national insights for different future scenarios, which was then

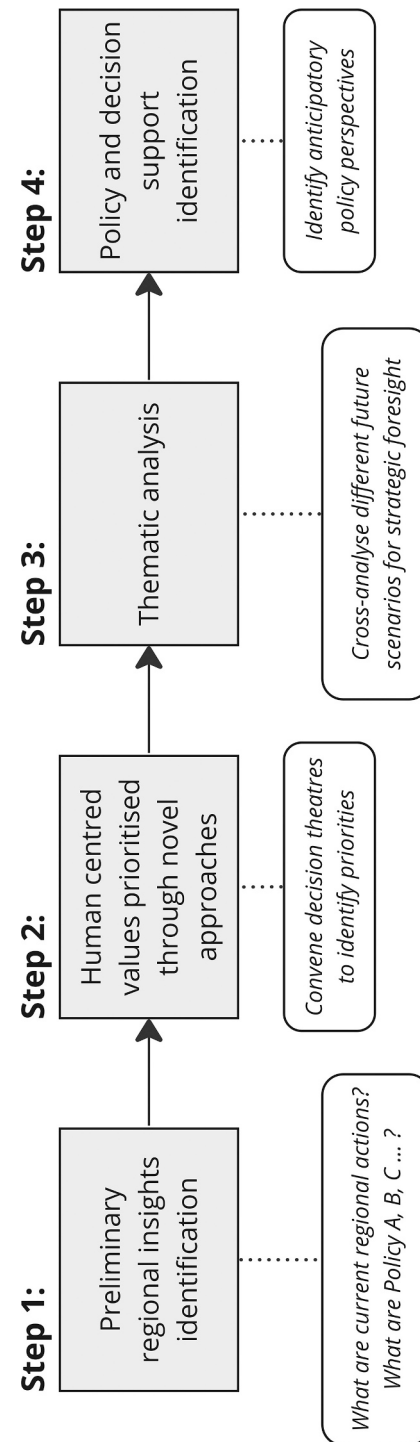


Fig. 2. The rapid policy analysis process as a research method.

analysed for policy opportunities. This dataset underwent thematic analysis [44] within this strategic approach, where priorities and concerns were classified, coded and clustered into themes.

4. **Policy support identification:** At this stage synergies and tensions between the regions were revealed. This comparative analysis exposed compatible and opposing priorities that were used to define anticipatory policy perspectives.

3.2. Creating the decision theatres

Most values-based insights are revealed in the second stage of the process - in the decision theatres. To simulate the decision theatres, UK and US workshops were held, with four sessions - two for each region. Participants were selected from interviewees in the first stage. They all had an intimate knowledge of CDR but from diverse perspectives. The aim of convening a disparate group of stakeholders was to create holistic discussion posited around technical and regulatory insights.

In these workshops, participants explored future actions and responses created when a hypothetical policy archetype, or strategy, was imposed nationally. Imposing a new archetype placed hypothetical pressure on the current policy regime allowing the stress testing of future responses to the current policy approach. Additionally, the hypothetical scenario and adoption of personas by participants created an environment for experts to reveal insights on challenges and opportunities without aligning closely to real events or internal biases, thereby producing more insightful outputs.

The archetypes used in this case study were adapted from the Vivid Economics 2019 Report on future policy pathways for CDR [45]. They offer pathways to achieve net zero by 2050 that integrate CDR. Two different policy archetypes were used in different workshops. The choice of two archetypes is to facilitate the strategic foresight approach. Multiple futures were created with each archetype and region combination. Then discussions from the four scenarios were cross-analysed and compared to reveal overlapping priorities, thereby creating anticipatory perspectives that are relevant in different future scenarios. The same two archetypes were tested in both regions to allow for regional comparison and reveal synergies and tensions. The archetypes were not used to recommend solutions for a CDR sector, as such, the intention was not to debate the merits or details of these policy options, but to understand the implications and likely response by different actors. The two policy archetypes used were:

1. **Upstream obligation approach:** Following the polluter pays principle, obligations are applied on upstream emitters, including suppliers of fossil fuels and agricultural products, and emitters with process emissions. 100% of the carbon content of the supplied products is balanced by an equivalent amount of CDR, with the proportion of removal increasing annually from now until 2050.
2. **Downstream price-led approach:** Downstream emitters can invest in CDR, through capital investment, or purchase of tradable credits, for certified tonnes of removal (\$/tCO₂). These are incentivised by tax credits. The value depends on the associated CDR option, which will have a price assigned to it according to factors such as cost, co-benefits and permanence. The prices will be reviewed annually.

The archetypes were discussed in two timeframes – the short term (the next 5 years) and the medium term (5–10 years) – with supplementary restrictions imposed at each stage to replicate likely future events. In the short term this included grants for commercialising solutions, like Biochar or DAC. In the medium term, this included expansion of the electric vehicle (EV) market, phase-down of Enhanced Oil Recovery (EOR), grants available for innovative CDR solutions, like Enhanced Weathering or Ocean Alkalinity Enhancement, and the availability of internationally tradable credits. The purpose of a richly impacted second phase was to target growing future concerns identified within the semi-structured interviews, including: the role of the oil and

Table 2

Personas used in workshops. Policy archetype 1 was used in UK1 and US1, archetype 2 was used in UK2 and US2.

Persona	Description	Decision theatre			
		UK1	UK2	US1	US2
Emitter: Hesitant to Engage	Large-scale emitter not yet comfortable engaging in CDR.	✓		✓	
Emitter: Transitioning Heavy Emitter	Corporate emitter with high embedded carbon content transitioning to sustainable business model.		✓		✓
Financial Investor	Investing in projects or companies with credible net zero plans.	✓		✓	✓
CDR Market Matchmaker	Actor in between CDR supply and demand.	✓			
Solution Supplier: Scaling Up	Supplier looking to commercialise (e.g. DAC).	✓	✓	✓	✓
Solution Supplier: Innovative Idea	Supplier transitioning from R&D phase (e.g. Enhanced Weathering).	✓	✓	✓	✓

gas industry in CDR's scaling; internationalisation of the CDR market; and a competing solutions space between CDR methods and markets.

A feature of the Decision Theatre Method [39] is allocating personas to participants, which they 'act' as during discussion. The function of the personas was to create objectivity and safe environments where participants are ensured confidentiality and trust, so as to not be restricted by personal or professional biases [46]. Participants were assigned a persona of a stakeholder in the CDR landscape that they understood and could speak for (this was established during the interview process). The personas used, summarised in Table 2, represent stakeholders that would be impacted by the imposed policy mechanisms but that can also influence their success. The personas were kept the same for both the UK and US workshops, to better establish similarities and tensions for comparison.

Using decision theatres, hypothetical archetypes and personas as tools in this research allowed identification of a broad set of priorities, generating exploratory and values-based insights.

3.3. Identifying policy insights

Discussion from the four decision theatres was transcribed before undergoing thematic analysis to reveal CDR policy priorities. To identify these priorities the discussions were coded [43]. This process involves pulling out relevant information, as it relates to the research aim of establishing a global CDR sector, into codes of opportunities, barriers and challenges. The coded results were grouped within the different decision theatre datasets. Clustering the findings allows priority areas to be identified. At this stage, key themes relevant to both regions and archetypes were identified, compared and analysed for priorities. With the same overarching themes identified, synergies and tensions were revealed in the results. From these results policy insights that deliver anticipatory perspectives to influence CDR development were identified.

4. Results and discussion: realising policy insights

This section outlines the key results from applying the rapid policy analysis process to UK and US CDR sectors. First the CDR policy landscape in both regions is mapped. Then the results from decision theatres are presented and analysed. Finally, the anticipatory policy perspectives outputted from this process are reported.

4.1. Mapping the regional CDR policy landscape

CDR methods have varying characteristics, including sequestration permanence, risks associated with development, removal potential, and readiness for immediate commercial deployment [5]. Thus, regions will pursue different CDR based on their localised priorities and resources. Closer examination of national CDR developments and values helps to reveal why different policies for CDR are being explored regionally, and consequently what CDR strategies have had the most success.

As step 1 in the rapid policy analysis process - see Fig. 2 - this subsection maps the CDR policy landscape. Mapping the landscape reveals existing regional priorities for UK and US policymakers, allowing for preliminary comparison.

4.1.1. UK policy

In 2020, the UK government published its Ten Point Plan for a Green Industrial Revolution, which laid out tangible actions that will be rolled out to achieve net zero [47]. Point 8 announced the use of £1 billion for 'Investing in Carbon Capture, Usage and Storage (CCUS)'. At the time this was the largest public commitment by a single nation to carbon capture and, although not directly contributing to CDR development, it demonstrates the UK's commitment to net zero. Since this time CDR policy mechanisms have emerged in the form of research funds, calls for evidence, incentives, codes and guidelines – the majority launched since 2020. These mechanisms are outlined in Table 3.

The UK government is exploring both nature-based and engineered methods at the national level – including a target of 5 MtCO₂ of engineered removal by 2030 [50] – building a CDR portfolio with total public funding across these strategies at over £2 billion. This portfolio is developing across multiple government departments, but with limited coordination. By developing programmes in isolation, developmental benefits – such as sharing knowledge, resources and experiences – cannot easily be reaped cross-departmentally.

The impact of this approach is being felt by industry early movers, with pressure for more government intervention. The Coalition for Negative Emissions, which represents organisations from traditionally hard-to-abate sectors that will require CDR to achieve their net zero targets, formed in 2020 to publicly recognise the critical importance of implementing CDR at national scale. The Coalition's 2022 solution-driven report [59] highlights that there is increasing demand for more targeted policy activities to facilitate successful scaling and address the government's current piecemeal approach. In recent years CDR in the UK has evolved from a research driven space to one of preliminary policy intervention.

4.1.2. US policy

A comparable policy picture can be mapped in the US. Carbon capture was first explored as a means to perpetuate the role of the oil and gas sector in a low-carbon world via Enhanced Oil Recovery (EOR) – a process that injects CO₂ into oilfields for easier extraction. Capturing carbon released from energy production and repurposing it for further oil extraction, omitting processes to chemically create new carbon, was the first use-case for US carbon capture. Although carbon removal is fundamentally different to carbon capture and storage (CCS), the overlapping technical needs of the two industries means CDR is often closely aligned to CCS. This highlights an added challenge for US CDR policy development: decoupling from strong fossil-fuel incumbent interests.

The US governance structure is more complex than the UK, with policy being implemented at both federal and state levels. Key mechanisms from both levels are outlined in Table 4.

Until recently federal CDR activity focused on two pathways: the 45Q tax credit for CCS in the oil and gas industry and management of nature-based removal. However, CDR policy mechanisms within the 2022 Inflation Reduction Act [60], 2022 CHIPS Act [61] and the 2021 Bipartisan Infrastructure Act [62], all launched under the Biden Administration, have provided a much-needed stimulus to the federal

Table 3

UK policy mechanisms or programmes for CDR (As of December 2022).

Programme or mechanism	Year	Department	Description
Greenhouse Gas Removal (GGR) Business Models – Consultation [48]	2022	Department for Business Energy and Industrial Strategy (BEIS)	Public consultation on GGR [CDR] business models conducted with aims to compile expert input on potential market frameworks (e.g. UK Emission Trading Scheme), policy design, monitoring, reporting and verification (MRV) approaches and wider insights. Public consultation seeking input on Bioenergy with Carbon Capture & Storage (BECCS) business models. BECCS is currently projected to have a large share of 2050 engineered removal targets in the UK's 6th Carbon Budget [50].
Business model for Power BECCS - Consultation [49]	2022	BEIS	DEFRA-BEIS collaboration with £12.5 m in funding to expand the evidence base for nature-based solutions, including Afforestation and Soil Carbon Sequestration.
Nature-based Solutions for Climate Change [51]	2021	Department for Environment Food and Rural Affairs (DEFRA)	An aim of the 25-year Environment Plan is the 'removal of greenhouse gases by forests'. To achieve this, grants of up to £100,000 are available through NEIRF fund for private nature-based projects.
Natural Environment Investment Readiness Fund (NEIRF) [52]	2021	DEFRA	£100 m in funding to find scalable engineered GGR [CDR] methods and projects, to achieve MtCO ₂ scale. 15 projects were selected for Phase 2 in July 2022.
DAC and GGR Innovation Programme [53]	2020	BEIS	Payment to landowners that are providing public goods, such as carbon sequestration. Indirectly supporting Soil Carbon Sequestration deployment.
Environment Land Management Scheme [54]	2020	DEFRA	Research assessing the economic infrastructure, market mechanisms and policy frameworks needed for engineered GGR [CDR] was commissioned by the UK's lead research body for infrastructure needs. Programme aiming to create six net zero industrial zones in the UK by 2040, alongside decarbonisation roadmaps, feasibility studies and supporting R&D. Two clusters to be operational by 2030.
GGR Technologies Study [55]	2020	National Infrastructure Commission (NIC)	
Industrial Clusters [56]	2020	UK Research & Innovation (UKRI)	

(continued on next page)

Table 3 (continued)

Programme or mechanism	Year	Department	Description
GGR Directorate Hub and Demonstrators [57]	2020	UKRI	Clusters are designed to include BECCS. Support five GGR [CDR] technologies to demonstration stage (GGR-D) and create a central hub (CO2RE) to coordinate and support market development. The hub is an Oxford-led consortium of academic experts.
Woodland Carbon Code [58]	2019	Forestry Commission	Credits available for woodland Afforestation until 2055. The incentive aims to scale nature-based removal.

CDR landscape. Additionally, congressional bills are increasingly introduced into the House [73,74,75], demonstrating heightened focus at the federal level. These new Bills and Acts are suggestive of a perspective shift towards a broader portfolio of technologies and policies across US government.

At the state level, momentum for CDR largely varies between states with California, a leader in climate action, currently acting as a key source for CDR policy. This is demonstrated by the \$200 per tonne Low Carbon Fuel Standard (LCFS) tax credit [72] accessible to developers across the country. Beyond California, state-led policies are being introduced in increasing number [76,77,78]. Passing these would increase the number of available incentives for private sector actors to utilise. By stacking state and federal policies the costs and risks of CDR deployment are decreasing, increasing developer opportunities and subsequent sector growth.

4.1.3. A fragmented policy landscape

This preliminary analysis of UK and US CDR policy highlights rapid development in the sector. The majority of policy support mapped has developed in the last five years alone. This quick rise has created a fragmentation of policies across departments. The UK and US have public finance and programmes accelerating across a range of technologies, and in multiple sectors, for example UK CDR policy has emerged from BEIS, DEFRA, the Forestry Commission, NIC and UKRI. Whilst technically exciting and demonstrative of increasing interest, policy development and subsequent fragmentation has occurred with an absence of national coordination, and even less internationally. In its current state, the tenants required for the development of a global sector will not be easily revealed.

This analysis has highlighted a fragmented landscape in need of better alignment, requiring a more systemic and coordinated approach than presently demonstrated to generate anticipatory policy design. This will enable CDR to effectively achieve economies of scale, reduce learning curves and overcome near-term scaling barriers.

4.2. Insights from the decision theatres

Step 2 of the rapid policy analysis process is convening the decision theatres, and step 3 is analysing the findings within a strategic approach. This subsection presents these results, first regionally then comparatively to reveal synergies and tensions.

Discussions from the decision theatres were coded to reveal themes that were relevant across the multiple scenarios explored by different policy archetypes and regions. This creates anticipatory insights that are relevant in multiple futures. The three themes identified in this process are as follows:

Table 4

US policy mechanisms or programmes for CDR. Federal Acts impose multiple mechanisms from one package out of the Executive Office. CDR mechanisms within these policy packages are outlined. (As of December 2022).

Programme or mechanism	Year	Department or State	Description
Federal Acts			
Inflation Reduction Act (IRA) [60]	2022	–	Includes updates to the 45Q tax credit, increases credits for bioenergy (impacting BECCS) and increases spending on rural areas (impacting nature-based CDR). Industrial and manufacturing funding package, focused on bolstering semiconductor leadership, includes \$1 billion for CDR RD&D. \$3.5 billion to develop 4 DAC hubs and additional funding for infrastructure development, such as CO ₂ transportation pipelines or injection wells. Key mechanism for agriculture and food policy that is updated every 5 years (next in 2023). It allocates RD&D funding through the USDA, relevant to biological CDR [64].
CHIPS Act [61]	2022	–	
Bipartisan Infrastructure Act [62]	2021	–	
Agriculture Improvement Act (Farm Act) [63]	2018	–	
Federal Policies			
Carbon Negative Shot [65]	2021	Department of Energy (DOE)	Mission aiming to accelerate breakthroughs for CDR through RD&D support, specifically engineered methods, with a goal to achieve \$100 per tCO ₂ by 2030. Provides technical and financial assistance to improve grazing conditions, increase crop resiliency, or develop wildlife habitat, increasing nature-based CDR.
Conservation Stewardship Program [66]	2021	United States Department of Agriculture (USDA)	Provides technical and financial assistance to agriculture producers and forest landowners. Land conservation incentives that support Afforestation and Soil Carbon Sequestration.
Environmental Quality Incentives Program [67]	2019	USDA	Initiative aims to address feasibility gaps in CCS, to identify safe storage and facilitate permitting, impacting BECCS and DAC.
CarbonSAFE Initiative [68]	2016	DOE	Tax credit first developed in 2008 for EOR and CCS. Recently updated in 2022 IRA. The credit was raised from \$50 to \$180 per tonne for DAC and a new minimum capacity of 1,000 tCO ₂ , down from 100,000 tCO ₂ , allowing small-scale developers to qualify.
45Q Carbon Tax Credit (Internal Revenue Code) [69]	2008	Treasury	

(continued on next page)

Table 4 (continued)

Programme or mechanism	Year	Department or State	Description
Renewable Fuel Standard [70]	2007	Environmental Protection Agency (EPA)	Created to boost renewable fuel sector with minimum standards on bio-based blends in transportation fuel. This supports biofuel and bioenergy industry, indirectly advancing BECCS and Biochar.
State Policies			
Carbon Dioxide Removal Market Development Act [71]	2022	California	Aims to develop and support CDR technologies, and has created a combined CDR and CCS target of 20 MtCO ₂ by 2030 and 100 MtCO ₂ by 2045 for California.
Low Carbon Fuel Standard [72]	2021	California	A regional regulated market with up to \$200 per tonne available for DAC, BECCS and CCS across the country.

1. **Economic mechanisms:** Financial capabilities, priorities and barriers
2. **Social contexts:** Human response priorities and barriers
3. **Technical characteristics:** Infrastructure and physical capabilities, priorities and barriers

Themes allow for comparative analysis between the two sectors. A limitation of this research identified in the thematic analysis process is the interrelated nature of the themes. Advancement in one theme can have an impact on another and therefore qualitative judgement is required to classify the findings into one main theme. The results from thematic analysis are summarised in Fig. 3.

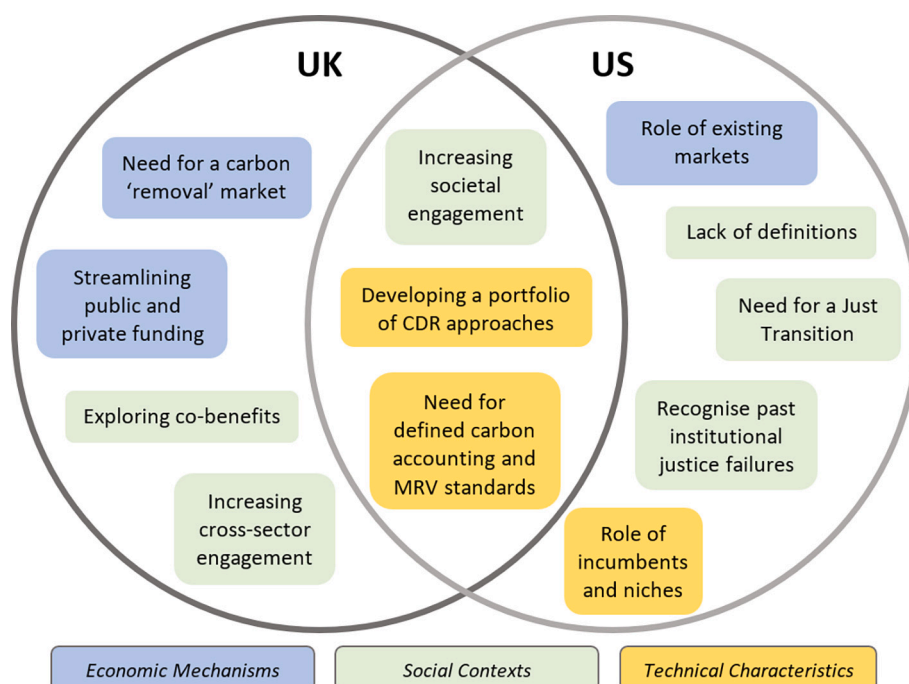


Fig. 3. Results: key thematic priorities for CDR development identified by UK and US stakeholders.

4.2.1. Priorities in the UK and US

Priorities differed between the regions. For example, in the UK a demand for cross-sector engagement was emphasised, the absence of which is seen as a barrier to growth. The fragmentation of policy across departments, as seen in Tables 3 and 4, is the likely cause of this engagement disconnect. In the US, discussions prioritised Environmental Justice (EJ) as a central tenant of all future CDR policy development which if insufficiently addressed could be restrictive to policy success. Discussion naturally prioritised regional issues, however, challenges identified are likely to be faced directly or indirectly in most regions during policy development in a global CDR sector. Tables 5 and 6 present a breakdown of the key regional findings.

4.2.2. Analysing synergies and tensions

Priorities for the CDR sector and subsequent policy development in the two regions will have similarities and key differences dependent on local values. To generate anticipatory perspectives, distinguishing these will be valuable. Key similarities between the UK and US include:

1. **A lack of societal engagement by both the public and policymakers:** Lack of public engagement is limiting growth in both regions. Also, despite the intersectionality of CDR, there is a lack of cross-department engagement by policymakers. Both stem from sector immaturity, lack of understanding and lack of strategic communication.
2. **An absence of consistent Monitoring, Reporting & Verification (MRV) standards:** MRV is currently insufficient or non-existent for CDR methods. Some methods are grouped into one methodology, which is an oversimplification. Lack of standards sees multiple private agencies providing disjointed systems of quantifying CDR and creating inconsistencies in measurements.
3. **A diverse CDR technology and policy portfolio:** Fragmentation demonstrates localised interests and priorities. This approach lends itself to diverse policy portfolio development, but fragments need to be nurtured through coordination and not abandoned.

Additionally, there were two similarities not explicitly prioritised in workshops but that were highlighted in the interview stage. First, the

Table 5
Key priorities revealed in UK workshops.

Theme	UK priority	Key insights
Economic mechanism	Carbon 'removal' market	• Need for a robust, adaptive and stabilising common carbon market that houses both removal and emissions reduction. • Accommodating to innovative technologies and projects of varying scale. • Lack of clear market increases risk as investment occurs without use of existing market tools and mechanisms for credibility.
	Funding streams	• Government support, specifically funding, is necessary in the near term. • Lack of CDR regulatory ecosystem calls for catalytic investment. • Lack of national understanding or awareness of necessity in net zero is prohibitive.
Social context	Societal engagement	• CDR needs acknowledgement as public good alongside abatement. • Current fears that CDR equates to business-as-usual for heavy polluting activity due to lack of engagement and understanding.
	Cross-sector engagement	• Currently there is a lack of awareness of nature-based policy by engineered CDR policy creators, and vice versa, despite a clear cross-cutting aspect of CDR. • Creates oversight of collaborative opportunities and learnings.
	Co-benefits	• Co-benefits, the benefits a policy or technology aimed at one goal could have on another, are currently overlooked. • Could be a tool to generate engagement, and economic and social benefit.
	Standards and metrics	• Current standards are catered to the existing avoidance-focused offset market. • Huge variability between solutions calls for immediate development of method-specific standards that are verifiable.
Technical characteristic	CDR portfolio	• Different approaches have different risks – support towards multiple methods and policy mechanisms is more suitable to mitigate this risk. • Prevents premature selection of a 'best' project or policy.

need for a stabilising carbon market that can manage avoidance and removal as well as successfully coexist with regulated markets, with potential for convergence, was prioritised in UK workshops and mentioned in several interviews with US stakeholders. Second, an underlying need to de-risk the entire CDR space to achieve near-term growth was highlighted in multiple interviews. This can be achieved by developing market tools in the CDR ecosystem that reduce risk and help projects reach demonstration.

Some challenges were prioritised in one region but not discussed by the other. These include:

1. **CDR funding streams (prioritised in UK):** National and private sector support varies between regions. Existing regional regulatory and financial mechanisms will dictate policy needs and must be understood for relevant intervention.
2. **The role of incumbent sectors or actors (prioritised in US):** Dominant sectors and markets vary between countries, some acting catalytically, some restrictively. For example, the oil and gas sector is less dominant in the UK than US, which will influence further support needs.
3. **Recognition of past failures in governance and Environmental Justice (EJ) (prioritised in US):** EJ was a priority in US workshops, stemming from growing momentum to address systemic governance

Table 6
Key priorities revealed in US workshops.

Theme	US priority	Key insights
Economic mechanism	Existing markets	• Existing voluntary and regulated markets need to be utilised for their tools and resources. • Entrance of engineered CDR may disrupt and stress markets, affecting credit pricing. • Absence of unanimous national definition is a risk, e.g. capture versus removal.
	Definitions	• Definition absence linked to nascent nature of sector and lack of national recognition of CDR's need.
	Societal engagement	• CDR is not understood as necessary, avoiding requirement to engage. • Public understanding and definitions for removal are connected. • History of racial and economic injustice in policy has created distrust and fear which inhibits engagement.
Social context	Environmental justice	• Some policy mechanisms (e.g. nature-based incentives) are exclusive to large landowners only, which are often corporate landowners. • Who is benefitting from policy needs to be monitored and addressed if necessary.
	Just Transition	• A Just Transition for all sectors and people is needed to prevent further marginalisation. • This means engaging with the industrial sectors and recognising a need for transparency in these relationships.
	Incumbents	• Incumbent oil and gas sector needs to be used positively in the net zero transition. It has a large influence in communities. • Expertise, resources and skillsets in oil and gas are applicable to CDR.
	Technical characteristic	• Building from incumbents requires monitoring to avoid greenwashing. • Absence of robust monitoring, reporting and verifying standards is limiting.
Technical characteristic	Standards and metrics	• Lack of emphasis on additionality and potential for double counting create risk. • Unifying standards could reduce inconsistencies between projects.
	CDR portfolio	• Policy developing at state and federal level is creating a portfolio. These policies need to be stackable. • Concerns that method selection is narrowing to one solution (e.g. DAC).

failures. Justice issues are not US-specific – they will reveal themselves differently based on localised circumstances. Top-down policy intervention risks overlooking local needs and justice priorities.

Regional synergies highlight opportunities where similar interventions or coordinated actions could be successful. Whereas tensions call for more targeted, region-specific support. Implementing a singular international policy strategy would oversimplify these tensions and create future roadblocks.

4.3. Framework of collaboration

The priorities outlined reveal common challenges facing and opportunities for CDR. By applying strategic foresight and human centred methods, the rapid policy analysis suggests these findings can be better coordinated under a globally collaborative approach. The concept of collaboration as a tool for improving CDR efficiency was introduced as a guiding principle by Honegger et al. [20]. The fragmentation within the

current CDR policy ecosystem, both at the regional and international scale, calls for better harmonisation - collaboration can address this.

To foster CDR to GtCO₂ scale in multiple possible futures, this research identified both international collaboration opportunities and national specialisations within a framework of collaboration. This approach develops upon existing regional policy mechanisms that have potential for success and initiates new ones under an overarching system of coordination. The framework identifies three opportunities for collaboration to align CDR policy fragments from the top-down. Then, to address regional variations in priorities and ensure nurturing and targeted support, identifies three national specialisations that promote action from the bottom-up. These are outlined in Table 7. The opportunities identified are near-term policy interventions, aiming to contend against the evolving nature of the CDR sector.

Table 7

Framework of collaboration: the rapid policy analysis process identified the following opportunities.

Opportunity	Description, reasoning and analogues
International collaboration	Instil CDR's role via a global roadmap that signposts globally agreed upon CDR capacity and infrastructure targets (e.g. Article 6 development or mechanisms needed for carbon market integration) that regions can contribute towards. Additionally, outlines safeguards to prevent CDR's implementation as an excuse for continued emissions.
CDR roadmap	- The IPCC's Special Report of 1.5 °C [2] was identified in this research as a key driver for CDR action to date. More can be achieved with CDR-specific guidance. - The International Energy Agency's (IEA) Roadmap for the Energy Sector [79] and the UK's 6th Carbon Budget [50] have begun discussing CDR in a roadmap context but via a singular 2050 target for a specific industry or region, respectively. They lack CDR-specific substance to catalyse effective action. This regulatory guidance would act similarly to the Montreal Protocol [80], London Convention [81] or Convention on Biological Diversity [82], that through international agreement successfully set clear guidelines for a specific technology set. - An internationally governed hub, that curates the above roadmap as well as monitors and coordinates regional CDR portfolios, acting as top-down advisory support to ensure roadmap alignment, overcoming limitations of communication in the current fragmentation.
Coordination hub	- This research revealed no coordination between the UK and US, and a lack of cross-department engagement in either region. There are opportunities to be reaped by sharing learnings and resources. A hub that can align technical niches would create more interconnected CDR mechanisms. - The International Space Station [83], and space travel more broadly, is an analogue of successful collaboration from a unified hub. International policy agreement and shared resources transitioned space travel from short term regional success to faster rates of global scientific and technological learnings. Central coordination of regional action could see similar success in CDR.
Agile, adaptive and resilient policy processes	- The final opportunity appeals for a philosophical development to accommodate for CDR's nascence and novelty. To manage uncertainties, early-stage governance cannot be stagnant nor rely on past circumstance. Rather it must be agile towards new policies, adaptive to future possibilities and resilient to changing circumstance. - Discussions highlighted an emerging transition from an extractive society to a regenerative one [84] in the net zero transition. CDR will likewise require a more resilient philosophical approach for implemented policies to remain applicable.
National specialisation	Policy fragmentation at the national level requires management. This can be achieved through regional CDR specific offices that coordinate activities, provide tools and resources, and act as an engaged local CDR governance. These offices would provide expertise and insights to nurture local priorities and issues from informed perspectives, and directly communicate and inform the global CDR hub.
Regional coordinating offices	- Research identified the emergence of CDR policy as demonstrative of growing interest, but concerns were raised over the effectiveness of the fragments without management. This increase in bottom-up communication would reduce oversight, knowledge gaps, ensure optimal resource use and work to safeguard equitable growth. - This form of coordination is emerging, such as through the UK's CO2RE Hub [57] or the US-based, method-centric DAC Coalition [85]. There is opportunity to utilise available knowledge and resources more effectively through structured public coordination from regional bodies.
MRV standards	- MRV methodologies provide credibility and de-risk the sector. Consistency in these standards will provide greater confidence in emergent methods, increasing engagement. These are important for integration into global carbon markets. In the near term these metrics need to be defined at a regional level. Technologies vary in maturity, storage permanence, and saturation capacity between regions. Defining standards locally will more accurately account for variations. As CDR scales these metrics will likely converge to a unified standard, aligning the quantifiable fragments. - The absence of consistent metrics was identified as a key challenge in both regions. Gaining public buy-in without demonstrating measurability success is challenging. - Nationally Determined Contributions (NDC), a required reporting commitment of the Paris Agreement, are exactly that – nationally determined. While these are claims and not metrics, they recognise regional circumstances in their baselines [86] but all act towards a common climate target. CDR needs similar regional systems for quantifying success. - Niche academic and scientific communities are cognisant of CDR's role. However, the wider public is uninformed – capacity needs to develop. Acting from solely the top-down without developing capacity from the bottom-up is limiting.
Public capacity building	- Lack of engagement stemming from lack of understanding was identified in both regions. - Building public capacity would connect the emergent and novel CDR sector to local values and priorities, reducing future equity challenges. Education is an effective, relatively low-touch tool for successful climate action [87,88] and a first step in public capacity building.

5. Conclusion & Further Research

CDR policy is a fast-evolving research area as a function of its growing importance in the net zero transition. In its current state the CDR policy ecosystem is nascent with technological deployment slow to emerge and an increasingly fragmented policy landscape. Traditional backwards looking, market-based, economically optimised policy insight generation tools can become quickly outdated in fast-moving sectors. Therefore, applying these tools to CDR policy design will be unable to generate necessary anticipatory insights in line with innovators needs. The rapid policy analysis process proposes an exploratory approach which better accommodates uncertainty, recognising that decisions on resource allocation are made by individuals. This process produces forward facing and values-based insights that engage with the multiple objectives inherent during innovating and scaling

novel technologies. Prototyping this process on the UK and US CDR sectors revealed anticipatory perspectives, highlighting collaboration as a guiding framework for this dynamic sector while valuing regional priorities through national specialisations.

Despite successful generation of anticipatory perspectives, there are limitations to the process which need to be addressed through further research. The application of the decision theatre method intends to eliminate bias. However, values-based insights identified in this method are subjective to the actors engaged. By focussing on the UK and US their respective governance structures will have influence over the global insights generated. To overcome limitations the rapid policy analysis process needs to be further applied to other geographical jurisdictions.

As actors and policies emerge in the scaling CDR ecosystem, this process needs to be applied and critiqued for its success in generating anticipatory policy insights. How the approach will accommodate the expanded range of actors requires testing. In the UK, the government's focus on business models [48,49] as the mechanism by which the CDR sector will be established makes this integrated strategic foresight and human centred design approach even more salient. Meanwhile in the US, the testing of mechanisms launched under the Inflation Reduction Act [60] will require adaptation of the approach. As more nations implement CDR strategy it can be argued that this novel approach is even more justified as a function of the lack of data in these regions where qualitative underpinnings of CDR sector development become more appropriate. In future iterations, the Dynamic Adaptive Pathways approach [89] could be used to introduce timesteps, further augmenting the anticipatory insights that the approach might generate. We welcome the adoption, testing, refinement and critique of this novel approach in the hope of enhancing responsive policy design across net zero transitions.

Declaration of competing interest

The authors certify that they have no affiliations with or involvement in any organisation or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript.

Data availability

No data was used for the research described in the article.

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